EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

Name: Sowbarniga A | Register Number: 241401104
Subject: Low Code Application Development using Zoho Creator

Aim

To implement and explore the security, access-control, data-sharing and governance features of the Insurance Claim Request Form application in Zoho Creator.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use. In an insurance office, for example, a claims intake executive needs to see the claim requests handled by the claims team but does not need to change the assessment or settlement details.

241401104

Application Used

The practical was carried out in Zoho Creator, where the security and governance features of the application editor were explored. The Insurance Claim Request Form application is designed for managing insurance claim requests, and the settings studied here (User Permissions, Roles and Data Sharing) are available in every Zoho Creator application. The screenshots in this record were captured in the sample application “My first Creator app” of the same account.

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Insurance Claim Request Form application. It is not a screenshot of the Zoho Creator interface.

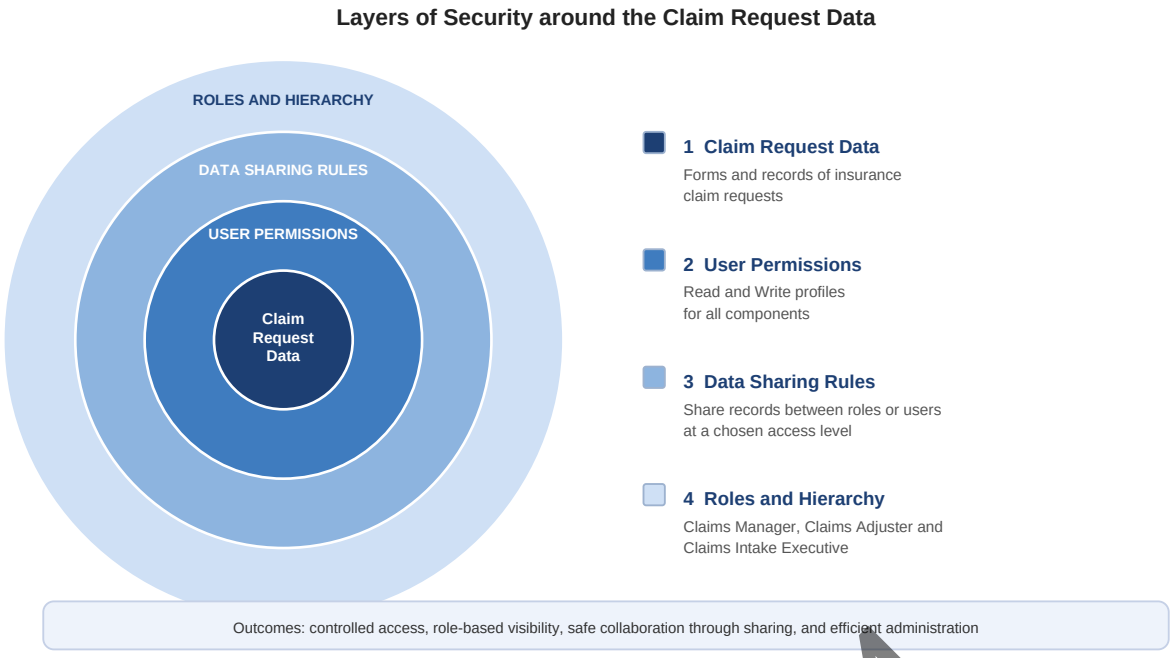


Figure 1: Security & Governance Layers (Conceptual Diagram)

The diagram shows the claim request data of the application at the core, protected by successive layers. User permissions (Read and Write) define what a user may do. Data sharing rules extend the visibility of records between roles or users at a chosen level of access. The role hierarchy forms the outermost layer and influences which records each role can see. The key outcomes are controlled access to data, role-based visibility, safe collaboration through data sharing, and efficient administration.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The application listed two permissions:

Name	Description (as displayed)
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components

241401104

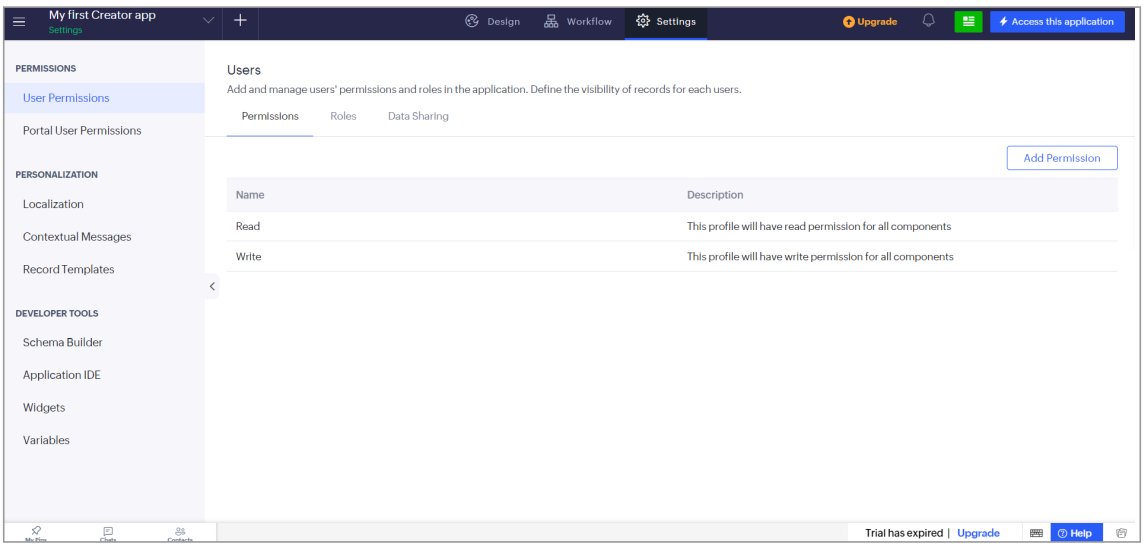


Figure 2: User Permissions (Permissions tab) in the Zoho Creator application editor

Read permission allows a user to view the components and records of the application without changing them, whereas Write permission allows the user to make changes, such as adding or modifying records. Further profiles, for example one for the claims adjusters of the insurance application, can be created with the Add Permission option; none was added in this experiment.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. In the account used, the tree contains the default top-level role, CEO. The options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available for building the hierarchy.

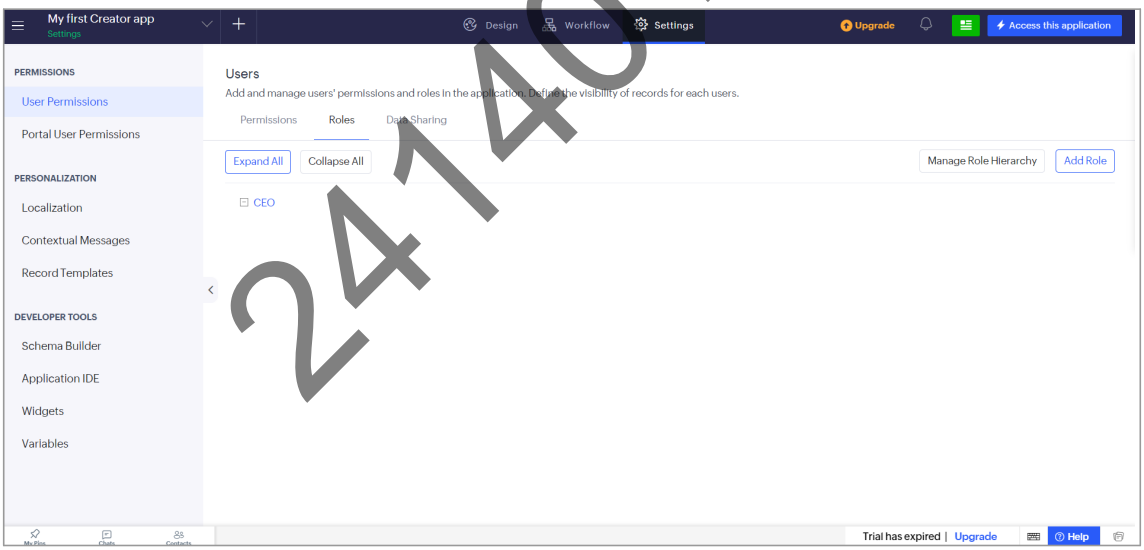


Figure 3: Roles and Role Hierarchy (Roles tab) in the Zoho Creator application editor

For the Insurance Claim Request Form application, the following role hierarchy is planned. The roles are added with Add Role and arranged with Manage Role Hierarchy; they are not yet created in the account.

Planned Role	Reports To	Position in Hierarchy
Claims Manager	–	Top level of the hierarchy
Claims Adjuster	Claims Manager	Second level
Claims Intake Executive	Claims Adjuster	Third level (lowest role)

241401104

Arranging roles in this way would give the application a clear reporting structure. The position of a role in the hierarchy influences which records its members can see, and further access can be given to a specific role through data sharing rules, as described in the next section.

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. The data sharing rule planned for the application has the following details:

Parameter	Planned Value
Rule Name	Claim_Request_Form
Form	Claim Request
Records Shared From	Roles – Claims Adjuster
Records Shared To	Roles – Claims Intake Executive
Level of Access	Read Only

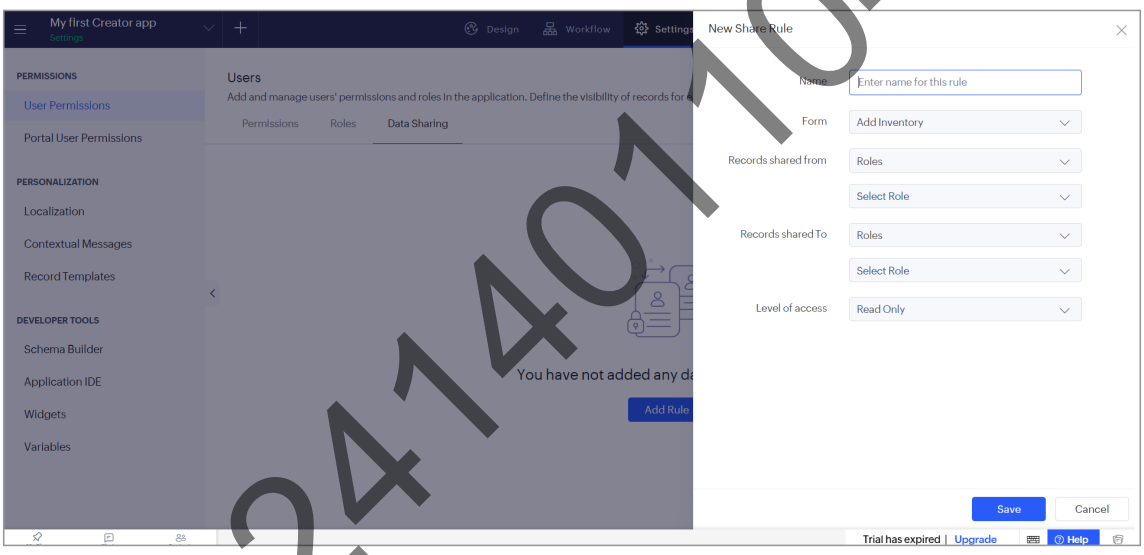


Figure 4: New Share Rule panel (Data Sharing tab) in the Zoho Creator application editor

Figure 4 shows the New Share Rule panel opened from the Data Sharing tab. The Data Sharing list behind it shows that no data sharing rule had been added yet. The panel has the fields Name, Form, Records shared from (with a role selector), Records shared To (with a role selector) and Level of access, which is Read Only by default. The rule was not saved in this experiment, so the table above describes the rule planned for the application once its roles exist.

The Records shared from field offers three choices: Roles, Roles and Subordinates, and Users. Roles shares the records of a single selected role, Roles and Subordinates shares the records of a role together with the roles beneath it in the hierarchy, and Users shares the records of selected individual users. With the planned rule, Roles would be selected, with the Claims Adjuster role as the source; with Read Only access, members of the Claims Intake Executive role could then view the shared records of the Claim Request form but could not modify them.

241401104

Security Features and Governance Controls

Figure 5 is a conceptual diagram that classifies the security and governance controls of Zoho Creator as they apply to the Insurance Claim Request Form application. It is not a screenshot, and it is not proof that every listed feature was configured.

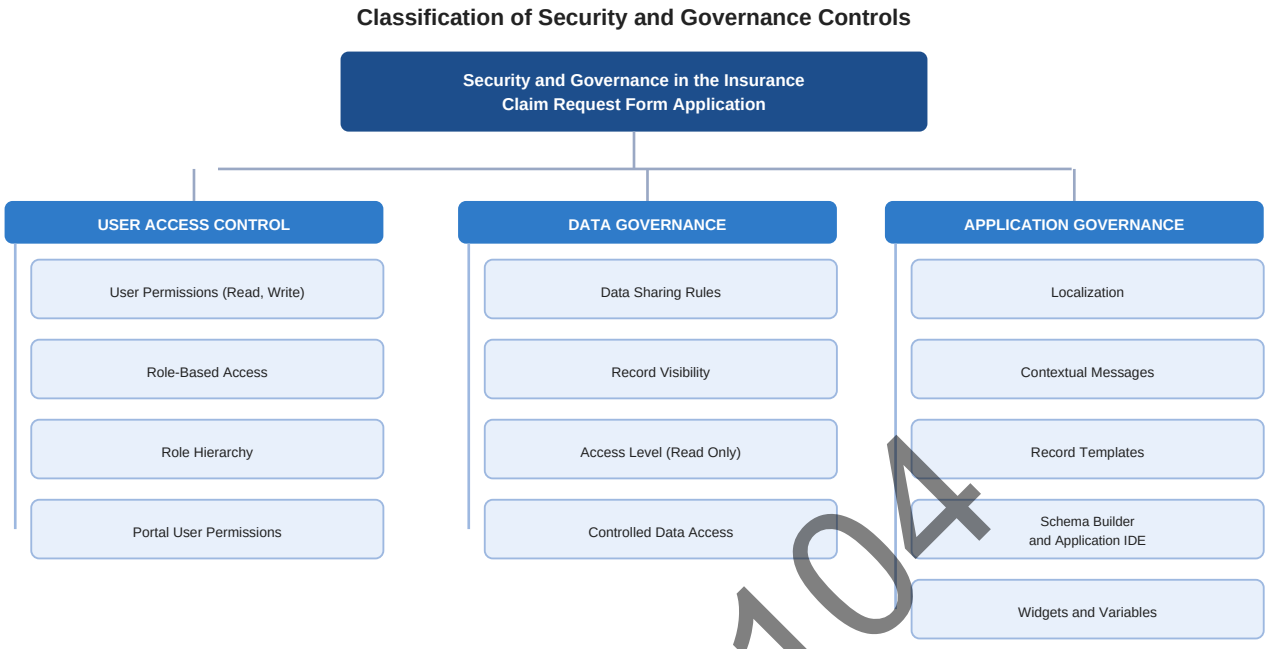


Figure 5: Classification of Security and Governance Controls (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read and Write), role-based access and portal user permissions determine who can use the application and what they can do.
Data governance	Data sharing, access level, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

In this practical, user permissions and roles were examined and the New Share Rule panel was opened. The other items in the diagram, such as portal user permissions, localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

241401104

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. It was not configured in this experiment, and no portal permissions were set for the Insurance Claim Request Form application.

Role-Based Access and Data Sharing Flow

Figure 6 is a conceptual diagram that combines the planned role hierarchy of the application with the planned data sharing rule. On the left, the roles Claims Manager, Claims Adjuster and Claims Intake Executive are shown in reporting order. On the right, the rule Claim_Request_Form is traced from the Claim Request form, through the role it is shared from, to the role it is shared to and the level of access given.

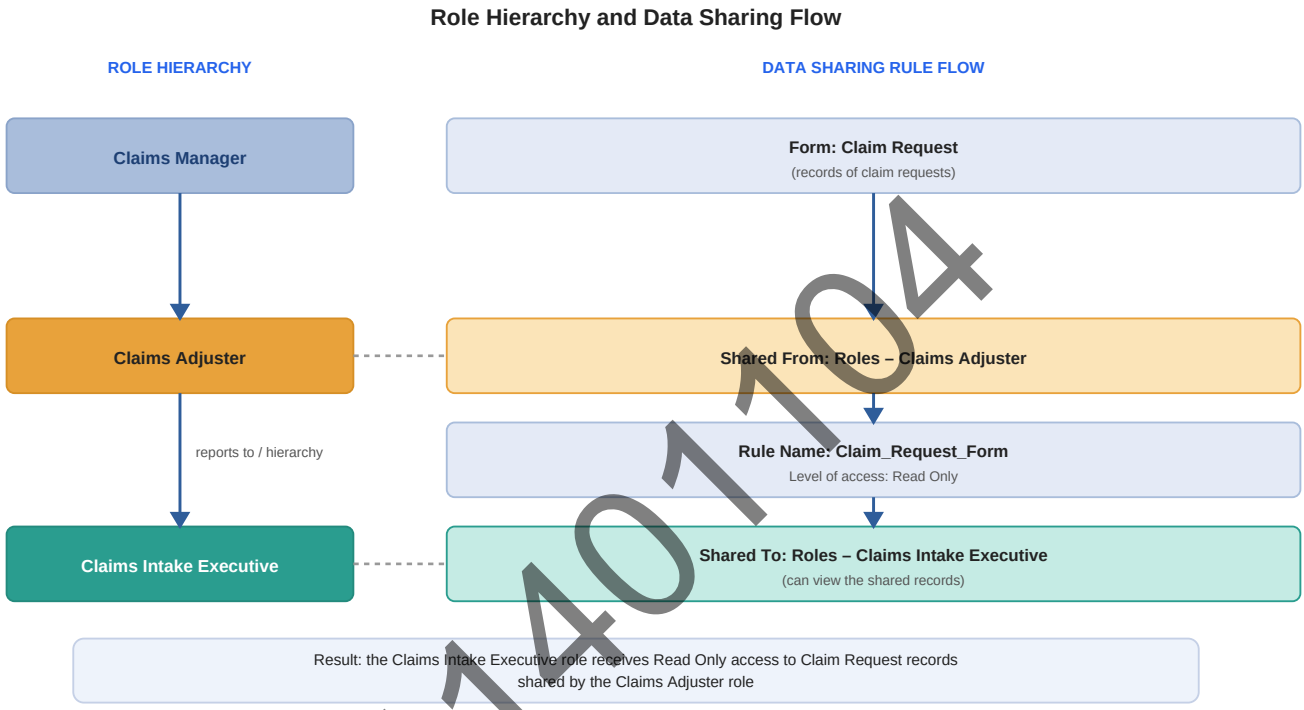


Figure 6: Role Hierarchy and Data Sharing Flow (Conceptual Diagram)

- The roles planned for the application form a hierarchy: Claims Manager, Claims Adjuster and Claims Intake Executive.
- The data sharing rule Claim_Request_Form is planned for the Claim Request form.
- Records would be shared from the Claims Adjuster role to the Claims Intake Executive role.
- The level of access is Read Only, so the Claims Intake Executive role could view the shared records but not modify them.

Governance Observations

- Predefined Read and Write permission profiles were available in the application.
- The Roles tab showed the default CEO role; the insurance claims roles are planned and are added with Add Role.
- The Data Sharing tab showed that no rule had been added, and the New Share Rule panel was opened to study its fields.
- The planned rule Claim_Request_Form shares records from the Claims Adjuster role to the Claims Intake Executive role with Read Only access.
- Portal User Permissions was not configured in this experiment.

241401104

- The practical showed how access can be limited to what a role needs by choosing a suitable level of access.

Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the Read and Write permission profiles were examined, the role tree of the application editor was viewed, and the New Share Rule panel of the Data Sharing tab was opened. A role hierarchy of Claims Manager, Claims Adjuster and Claims Intake Executive and a data sharing rule named Claim_Request_Form, sharing Claim Request records from the Claims Adjuster role to the Claims Intake Executive role with Read Only access, were planned for the Insurance Claim Request Form application.

The experiment shows how permissions, roles and data sharing work together so that each role sees the records it needs and no more. Choosing a suitable level of access, such as Read Only, allows records to be shared for collaboration without allowing them to be changed. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.

241401104